



ANBALAGAN LAKSHMANAN

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OBJECTIVE

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Seeking a challenging position as a **Lead Mobile Developer** to utilize my skills and abilities in the IT industry that offers professional growth while being resourceful, innovative, and flexible.

SKILLS SUMMARY

- Over 12+ years of Hands-on experience with **Native Android Mobile Applications design & development** using Java1.8, Android13.x using MVC, MVP, MVI, MVVM, Viper architectures, LiveData and Room DB.
- Over 5+ years of Hands-on experience with Kotlin1.8, KMM-Kotlin Multiplatform **Mobile**, StateFlow, Clean **Architecture**, Coroutines concurrency **design** pattern, JETPACK Compose, Glance, Work manager, Koin and Hilt dependency injections, multithreading, migration, interfacing, routing, **testing** and production support.
- 6+ Years' experience in Managing Android and IOS **Software development** teams, code reviews, resolve code merge conflicts, deployment, and post-launch monitoring as a **Software Development Lead**.
- Extensive experience in NAAAS-Native apps as a Service to developed Android different Audio and Video streaming **applications** and integrated into ARM64, Q845 and x86_64 hardware Platforms.
- Expertise and experiences in developing Android Data Streaming, Real time messaging and Ultra Low Latency.
- Extensive experience in Adding 3rd party production studio and cameras into Interactive Live streams.
- 5+ experiences in **design** and develop fastest, highest quality livestreams Android SDK/NDK Echo systems with Poll Live streaming Audiences and display results in real time.
- Experience in developing Security F/Ws to protect customers data using SSL, TLS, DTLS with 256 encryptions.
- Expertise and 12+ years experiences in Agile CI/CD process using GIT Jira, Jenkins/Bamboo/**Azure** & Bitbucket.
- In-depth knowledge of Kotlin1.7.0 and Android12.x UI **design** with Fluid UI, Zeplin, XD Adobe and Dynamics layout using Fragment Manager, ConstraintLayout, RecyclerView and Landscape and Portrait mode Processing.
- Experience in Performance **Testing** Tools includes Android Studio Profilers, Lint analyzer, System Tracing, Microbenchmark, Macro Benchmark, JUnit4, AndroidJUnitRunner, and Espresso.
- Experts in Android Java projects bugs finding using GitHub, Segment and Amplitude **analysis** Tool.
- Experience in Java **API** programming, **web services** JSON & REST **API development** for Android **applications**.
- Experiences in Android12.x application crash reporting solutions using Firebase Crashlytics, Lint Analyzer
- Expertise in MDM-**Mobile** device management Tools Integration and deployment with SOTI-Android Kandji-IOS.
- In-depth knowledge of Migrating from Android LiveData and Kotlin StateFlow with Room DB.
- Designed and Developed Android 12.x apps for Lexus IVI systems apps using Android Automotive Operating Systems (AAOS) and Google Automotive **Services** (GAS) with ALEXA SDK and Google Home SDK integrations.
- Worked and Developed Android Automotive OS apps (#cars #apps), Google Automotive **Services** (GAS), Android Core OS/Frameworks (#batterylife), Android ecosystem (#growth, #ecosystem, #security, #updates), **Cloud Services** for Android Partners (partner.android.com/home) (#b2b, #enterprise, #analytics) and Google App Distribution (#distribution).
- Experience in Android Repository CVS and distributed Version Control System using SVN, GIT and Clear Case.



- Expertise in building, installing, configuring, compiling, shrinking and debugging the Linux kernel modules using U-boot, MTD Partitions, ram disk and NFS file system for enterprise network applications.
- Sound knowledge in Linux Busy box, Cross Compiler, libraries (libc), building manual and automatic apps.

EXPERTISE

Programming, Cloud & Agile Experience:

- Java1.8, Python 3.11.3, Go language
- REST API: Postman, Swagger, JWT.io
- Build Tools : Gradle1.11, Maven
- CD/CI: Jenkins, Azure, AWS Bamboo
- Repo: Bitbucket, GitHub, GIT, SourceTree
- MDM: SOTI-Android and Kandji-IOS
- Agile: JIRA, Confluence, Bitbucket,
- OCR: Micro Blink, IDOLOGY, Mi Snap
- Payment FWs: Braintree, Stripe, Picasso

Cloud (AWS, Azure, Google) Experience:

- AWS, Azure, ForgeRock and GCP
- AWS Voice: AMAZON Speech Recognition
- AWS: S3, SNS, Pinpoint, Transcribe
- ForgeRock Identity Cloud and Platform
- Voice Integrations: Google, NUANCE F/w
- Azure – Boards, Repos, Pipelines

Kotlin, Android F/Ws, & API Experience:

- Kotlin1.8, Kotlin-KMM, Coroutines, JETPACK
- Android13.x, Android Studio3.2, AOSP, Profile
- JETPACK: Compose, Glance, Work Manager
- MVVM, MVC, MVP, MVI, Viper, Clean Architecture
- Android – LiveData, Kotlin – StateFlow, Butterknife
- Dep. Injections: Koin, Hilt, Dagger, Retrofit, RxJava
- F/Ws: React Native, Flutter, Ionic, NativeScript
- Analytics: mParticle, Amplitude, Segment, Lint
- BDD & TDD Tools : Airborne, Espresso, Cucumber

Android OS and NDK Experiences:

- Architectures: ARM32, ARM64, x86 and x86_64
- OS : Android 14, Linux Kernel5.x and 6.1
- Hardware : Qualcomm845, OMAP, Ras Pi-4
- Cross Platform Tools : Vulkan, Vulkan SC & API
- Android Base Profile: 2021, 2022 & Vulkan1.1
- Build Tools : – Cygwin, Cmake, Eclipse

PROFESSIONAL EXPERIENCE

Newway Healthcare Services, Upper Darby, PA

Aug 2022 – Present

As a Software Development Lead

- Built and Integrated Android Audio and Video streaming Libraries Kagiso & FFMPEG into ARM64 & x86_64 Platforms.
- Designed & Developed Native Android13.x & Kotlin1.8.0 LiveSwitch Medwand Mobile Clinic Chat Application using Android AOSP libraries for HP Engage Go 10 Mobile systems and Intel x86_64 Platforms.
- Building and Recompiling libopus.a, libopus.so, libvpx.a, libvpx.so, libyuv.a and libyuv.so for Android AOSP x86_64 Platforms <https://chromium.googlesource.com/libyuv/libyuv/>.
- Integrated and successfully installed all the libxxx.a and libxxx.so libraries into HP Engage Go 10 Mobile systems.

PROJECTS HANDLED:

- Developed and released LiveSwitch Medwand Mobile Clinic Chat Application using Android13.x and Android AOSP.
- App : <https://www.medwandhealth.com/products>

Environment:

- OS : Android14, Linux Kernel 5.15
- HW Platforms : ARM64 and x86_64
- NDK Build Tools : CMAKE, Cygwin
- Audio/Video Stream Library : Kagiso, FFMPEG
- JETPACK: Compose, Glance and Coroutines
- CD/CI: JIRA, Jenkins, Bitbucket, GitHub



- MDM: SOTI-Android and Kandji-IOS
- MVVM, MVC, MVI, Clean **Architecture**
- LiveData, StateFloww, RxJava3, Retrofit2, Room DB
- **Azure** – Boards, Repos, Pipelines, Test Plans

HerringBank, Amerillo, TX

Jan 2020 – Jul 2022

As a **Software Development Manager**

- Designed & Developed **Native Android13.x & Kotlin1.8.0** **Mobile** Wallet Application & Merchant POS Application.
- Integrated IDOLOGY OCR Scan Verify SDK is a service that ability to capture and validate Driving License Photo ID.
- Integrated IDOLOGY OCR Scan onboard **API** is a microservices to send consumer front and back DL images to IDOLOGY server and Server will authenticate and authorize the consumer registration.
- Integrated MicroBlink SDK to scan the credit card and call the **API** to add credit card as Funding Sources.
- Integrated Pay-in-Store, ADD Fund, Send Money Rest **API** calls with Retrofit, RxJava, and OKHttp.

PROJECTS HANDLED:

- Developed and released HerringBank **Mobile** Wallet Application using Idology OCR and Android12.x.
- Developed and released HerringBank Merchant POS application using QR code scanner and Android12.x.
- App : <https://play.google.com/store/apps/details?id=com.apiture.xpressmobile.hb3atx.sub>

Environment:

- Java8.0, Kotlin1.8-KMM, Android13.x
- MVVM, MVC, MVI, Clean **Architecture**
- OCR: MicroBlink, IDOLOGY, MiSnap
- Android – LiveData, Kotlin – StateFlow
- Firebase msg, Twilio, **Amazon** Pinpoint
- MDM: SOTI-Android and Kandji-IOS
- JETPACK: Compose, Glance and Coroutines
- Koin, Hilt, RxJava3, Retrofit2, Butterknife, Dagger
- App Center, TestFlight, Fastlane, CocoaPods, Junit
- CD/CI: JIRA, Jenkins, Bitbucket, GitHub and Git
- Sodium Lib, GLIDE, Picasso, Espresso, and Zxing Lib
- **Azure** – Boards, Repos, Pipelines, Test Plans

Tuya Technology, Houston, TX

June 2019 – JAN 2020

As a Engineering Manager - Android

- Developed **Native Android 10.x** Tuya tech Driver and Shipper **applications** using Google Maps, Room DB.
- Integrated Driver on-onboarding and background checks frameworks, Fountain **API**'s, Turn.auth, and Truckright.

PROJECTS HANDLED:

- Developed and released Tuya Tech Android12.x Driver app using Room DB and Fountain SDK.
- Developed and released Tuya Tech Android12.x Shipper app using Room DB and Fountain SDK.
- Shipper app : <https://apps.apple.com/us/app/tuya-shipper/id1473548878>

Environment:

- Java8.0 Programming, JIRA, Jenkins
- Retrofit2.x, Butterknife, Dagger, OKHttp
- Firebase messaging, **Amazon** Pinpoint
- **Microsoft Azure**, App Center, Bitbucket
- MDM: SOTI-Android and Kandji-IOS
- Android12.x, Android Studio3.4.1, Kotlin1.3.11
- JETPACK Compose, Glance, Hilt, Koin, Room DB
- Fountain, Turn.auth, Truckright, Zeplin.io, Swagger
- Google Maps, Room DB, Firebase, Glide
- **Azure** – Boards, Repos, Pipelines, Test Plans

Lenovo, North Carolina, NC

Nov 2018 – May 2019

As a Lead Android **Developer**

- Designed & Developed **Native Android 9.x** **Lenovo think Reality** AR/VR launcher APP & Settings App. using AOSP.
- Modified system level Android11.x AOSP frameworks and Build, Flash and integrated in to Lenovo Think Reality H/W.



- Designed, Developed and Modified Google Android 9.x AOSP frameworks and setting library into **Lenovo** Think Reality 2D and 3D Android11.x frameworks and Layouts. .

PROJECTS HANDLED:

- Developed **Lenovo** Think Reality Android 9.x 2D **Mobile** app with Android AOSP New settings features.
- Developed **Lenovo** Think Reality Service security app to with Hardware level Android notifications.
- <https://techtoday.lenovo.com/us/en/solutions/thinkreality>

Environment:

- Java8.0 Programming, JIRA, Jenkins
- Retrofit2.x, Butterknife, Dagger, OKHttp
- RxJava, RxBindings2.x, RxPermiss2.x
- Bitbucket, Bamboo, GIT, Jira
- Android11.x, Android Studio3.1.x, Kotlin1.x.x
- Swagger, Zeplin.io, Gradle3.45.x, **Apache** Maven
- Android Extension Libraries (AndroidX), Jetpack
- Lenovo Think Reality HW and Qualcomm 845

Toyota Motors North America, Plano, TX

May 2018 – Nov 2018

As a Lead Android **Developer**

- Worked and Developed Android Automotive OS apps (#cars #apps), Google Automotive **Services** (GAS), Android Core OS/Frameworks (#batterylife), Android ecosystem (#growth, #ecosystem), devices secure (#security, #updates), **Cloud Services** for Android Partners (partner.android.com/home) (#b2b, #enterprise, #analytics) and Google App Distribution (#distribution).
- Designed and Developed Android 12.x apps for Lexus IVI systems apps using Android Automotive Operating Systems (AAOS) and Google Automotive **Services** (GAS) with ALEXA SDK and Google Home SDK integrations.
- Designed and Developed Android 9.x frameworks and UI wireframes for Toyota and Lexus Android **mobile** application Login Portal, Subscription maintenance, Preferred Dealer portal, Dashboard, Check Engine portals.
- Integrated **Microsoft** MSAL **Azure** and ForgeRock OAuth2.x-OpenID-auth frameworks in Toyota **mobile** application.
- Integrated B4VScanner VIN number scanner SDK in to Toyota and Lexus 2018 vehicle Kotlin **Applications**.
- Developer** Dashboard using material **design** with CAN Bus integration, Series XM **API**'s & Driver score, Last Trip info.
- Developed Subscriptions maintenance with scheduling appointment with LBS location-based **services**.

PROJECTS HANDLED:

- Developed **Toyota** ONEAPP Android 9.x **Mobile** application to releasing on 2019 and 2020 cars.
- Developed **Lexus** ONEAPP Android 9.x **Mobile** application to releasing on 2019 and 2020 cars.
- Oneapp : <https://play.google.com/store/apps/details?id=com.toyota.oneapp>

Environment:

- Java8.0 Programming, JIRA, Jenkins
- AAOS: batterylife, chargepoint, car apps
- Android Auto: AAOS, CAS, IVI systems
- Google Fire-base, AWS SNS, Pinpoint
- Android11.x, Kotlin, Android Studio3.1.x
- ALEXA SDK and Google Home SDK
- VIN / QR Scanner: Bees4Honey, JSON, Rest APIs
- Oaths 2.x, OpenID, ForgeRock, MSAL **Azure**

Bank of America, Plano, TX

Aug 2016 – May 2018

As a Vice President (Principal Engineering Manager - **Mobile**)

- Implemented Android7.0 Libraries and Frameworks with Mastercard Masterpass **API**'s to BOFA **mobile** application.
- Integrated NUANCE Voice frameworks and **API**'s into all current Bank of America Android **Applications**.
- Developed Android7.0 Libraries and Frameworks with BOFA Small Business **API** into BOFA **mobile** application.
- Developing and implementing new business features with Android7.x and Google pixel devices.

PROJECTS HANDLED:

- Developed and Published Android7.1 BOFA **mobile** application with Mastercard Masterpass features.
- Developed Android7.1 NFC Application running on small business Payment and provisioning Flow & process.



- App : <https://play.google.com/store/apps/details?id=com.infonow.bofa>

Environment:

- Java8.0 Programming, JIRA, Jenkins
- Retrofit2.x, Butterknife, Dagger, OKHttp
- Android Multi window support
- Voice Integrations: NUANCE framework
- Gradle2.x, Cucumber BDD, Espresso
- Android10.0, Kotlin, Android Studio2.2.x
- RxJava, RxBindings2.x, RxPermissions2.x
- Glide, GSON, Braintree, Stripe, JSON, Push Notifi
- Analytics: mParticle, Amplitude, Segment, Lint
- Image Loaders FWs: Picasso, Fresco

Aditi Staffing, Santa Clara, CA

June 2015 – July 2016

As a Software Engineer – Android III (Client – INTEL, Santa Clara, CA)

INTEL True VR: Includes Intel VR headset, Intel True VR Camera SDK and True VR Android **Mobile** application that was part of Intel based Android devices. Intel True VR turns sports into amazing immersive experiences features.

Environment: Android Studio, Eclipse, ADB, python, Rally, Irfan view, phone flash tool

Google play Link: <https://apk.support/app/com.vokevvr.vokeviewer.daydream>

INTEL VR Commercial Applications: Includes Intel VR headset, Intel VR Camera SDK and VR Android **Mobile** application for retail, **Architecture**, Medical and educations that was part of Intel based Android devices.

Environment: Android Studio, Eclipse, ADB, python, Rally, Irfan view, phone flash tool

Depth Camera Applications: Includes Camera and Gallery **applications** that was part of Intel based Android devices. It had the capability of creating depth image and showcasing the depth features.

Environment: ADB, python, Rally, Irfan view, phone flash tool

- Designed and Developed 2D INTEL VR commercial **Applications** running on Android6.x and Android Studio3.x
- Designed and Developed 2D Depth camera **applications** and INTEL True VR **applications**.
- Working experiences with Intel DS4 Cameras and Fish eye cameras from WW10.xx to WW50.xx in all level of product **development** and integration with different Google requirement of Tango HAL and Tango CORE libs.
- Implemented Intel Real sense Android5.1 3D Camera Left, Right, Depth, RGB **API's** in to Intel Tango Device.
- Integrated Intel Android5.1 Depth and Fisheye motion camera **API's** in to Google Tango Frameworks.
- Implemented Tango HAL low level Libraries and **API's** in to Intel Android5.1 Tango devices.
- Integrated Tango Mapper Android calibration binaries and scripts file in to Intel Android5.1 Tango devices.
- Calibrated Intel tango device with Google MeasureIT and AndroidviCalib **Applications**.
- Integrated Android Compatibility Test Suits (CTS) in to Intel Tango Mapper Devices.

PROJECTS HANDLED:

- Developed Point **cloud** 3D Android5.1 Application using Intel Realsense SDK and Android5.1 framework.
- Developed Depth camera Android5.1 Application using Intel Realsense SDK and Android5.1 Tango devices.

Environment:

- Intel DS4 and Fish eye cameras
- Apps: Box measurement, point **cloud**
- Android NDK10.r, C, C++ and OpenCV
- Intel x86 Fisheye & Depth camera APIs
- Google Analytics, Google Ad Mob
- Android6.0, Android5.0, Intel Realsense SDK
- Tango mapper, Tango Explorer, GIT, Gerrit server
- Tango HAL Libraries and Tango HAL CTS
- Git/Gerrit/Repo, JIRA, TeamForge, Syncplicity
- Android CTS , Phone Flash Tool, DDMS, ADB-shell

TCS, Phoenix, AZ

June 2014 – June 2015

As an Application Architect (Client – American Express)



- Implemented Android4.4 NFC provisioning flow and Payment flow UI in American Express NFC Native application.
- Implemented Android Model-View-Controller(MVC) **design** pattern in American Express **Mobile** Application.
- Implemented Android5.0 Push Notifications **API's** in American Express receipt match Application.
- Integrating Fingerprint Scanner **API's** to NFC Payment Application and developing Custom Permission, Security Protocols to isolating the payment application from "Wood Glue Spoo" hackers.
- Debugging in Android APK Dex 65K method and Shrinking Google play **services** and Google Drive **API's**.

PROJECTS HANDLED:

- Developed American Express NFC Payment Native application using HCE AAR Libs running in Android Studio1.0.
- Developed American Express **Mobile** Wallet Application using Android Studio1.0 to UK and Australians market.
- APP : <https://play.google.com/store/apps/details?id=com.americanexpress.android.acctsvcs.us>

Environment:

- Android Studio1.0, Gradle2.1
- ProGuard
- Android 5.0, Android 4.4, AOSP
- Android5.0 Push Notifications **API's**
- Google Ad Mob and Analytics, Jenkins
- NFC HCE **API's** and Android Archive Library (AAR)
- Dagger, Butter Knife, GSON, Retrofit, RxJava
- OkHTTP Client, **Web Services** REST and JSON
- FingerPrint **API's**, Zipalign, Jarsigner, MobileIron

Park **Computer Systems, Newark, CA & Freelance Android Developer**

Nov 2013 – June 2014

As a Senior Android **Developer (Clients – Deutsche Telekom & SensoPlex, CA)**

- Implemented Android4.4 KitKat framework library using Low power sensor batching **API's**.
- Involved in Android full-screen immersive mode to supports Android 4.4 new gesture.
- Involved in data storing implementation between Android 4.4 sensors Google drive API2 SDK.
- **Design** and **development** with Android4.4 Fragment Manger.
- Right after my Deutsche Telekom project, I have started working as a Freelance Android **developer**.
- Involved in Android4.4 Blue tooth Low Energy (BLE) SDK and Android Application **development** for Sensoplex Fitness & Wellness Tracking module.
- Involved in Android4.4 NFC HCE Host based Card Emulation with PayPal CardIONFC **API** and Android Application **development**.
- Involved in Android4.4 Shared SMS Provider with new semantics **API's** and SMS MMS messenger application **development**.

PROJECTS HANDLED:

- Developed Android California Earth Quake Monitoring systems in Samsung S4.
- Developed Android framework Library to Accelerometer, Magnetometer and LBS **services**.
- Developed Android4.4 BLE Fall detection and prevention Bluetooth4.1 Wristband algorithms and application.
- Developed Android4.4 SMS MMS messenger new semantics and NFC HCE Host based Card Emulation **Applications**.

Environment:

- Google Drive API2 and Google Nexus 4
- Google Ad Mob, Google Analytics
- Android4.4, Bluetooth HID over GATT
- Android **Web services** REST and JSON
- Android 4.4 **API's** and SDK Samsung S4
- Android Location Based **Services API's**, GitHub
- Android 4.4 NFC HCE and new semantics **API's**
- Sensoplex Bluetooth LE smart sensor module

eTouch Systems, Fremont, CA

Sep 2013 – Nov 2013



As a Senior Android Developer (Client – Google Inc. Mountain View, CA)

- Implemented JSON and REST interface between Enterprise Launcher Application and Google application server.
- Involved in Android Voice recognition application framework development to Google nexus Tab and Nexus 4.
- Involved in Google drive API2 SDK integration in Launcher Application using Gmail ID.
- Involved in Android switch launcher application API development using Android LBS services.

PROJECTS HANDLED:

- Developed Android Enterprise Launcher Application to Google Nexus 4 and Samsung S4.
- Developed Android framework Library to access data from Google drive using REST and JSON protocol.
- Developed Android Enterprise Launcher Voice recognition application using LBS services API.

Environment:

- Google Drive API2 and Google Nexus 4
- Android Web services REST and JSON
- Google Enterprise Launcher API's
- Intel True VR and commercial Apps
- Android Voice Recognition API and SDK
- Android Location Based Services API's, GitHub
- Android JB4.2, Android ICS4.0, Samsung S4
- Intel Depth Camera Applications

Datacard Group, San Jose, CA

Apr 2013 – Sep 2013

As a Senior Android Developer

- Involved in Mobile wallet Architect framework development. Technical specification for Android, IOS, Black Berry.
- Involved in Android Mobile Wallet APDU communication protocol between Secure Elements NFC and μ SD.
- Involved in Android Mobile wallet card down load library between client Application and Datacard TSM.
- Involved in Mobile Wallet device eligibility library framework to S4 between DataCard TSM and MNO AT&T.
- Implemented RAM over HTTP request communication for Mobile Network Operator(MNO) and Service Providers.

PROJECTS HANDLED:

- Developed Android Mobile Wallet Library Framework and Platforms development and deployment to Banks and SP's.
- Developed Android Mobile Wallet demo application to Datacard Group and deployed Samsung S4 and μ SD.
- Developed Phonegap application to build cross platform Android, IOS, BB using command line interface.

Environment:

- PhoneGap API's, Cordova, CLI, GitHub
- Android Web services REST and JSON
- Android fragments, AIDL, Binder IPC
- Android JB4.2, Android ICS4.0, Samsung S4
- RAM over HTTP request and APDU protocol, NFC HCE
- ISIS Wallet, Nexus 4, NFC, μ SD, MNO, TSM, SP

OKAYA Inc, Centereach, NY

Mar 2012 – Jan 2013

As an Android Software Programmer – (Client – INTEL, Santa Clara, CA)

- Implemented Android web client applications and web services REST & JSON in Intel Android ICS phones.
- Involved in developing Geo fence application frame work to INTEL mobile using Location Based Services.
- Implemented Java Simple and Complex Context Source in INTEL Midfield Phone using Android ICS NDK.
- Implemented Facebook Android SDK and Graph API in Android4.0 ICE Cream Sandwich INTEL Midfield Phone.
- Involved in Android OpenCV and open source 3D video algorithms porting on Intel phone Midfield PR3x – ICS4.x.

PROJECTS HANDLED:



- Developed Location based **services** Android apps Locale3.x and Gimbal4.0 to Intel Android ICS phones.
- Developed Geo fence application using Android4.0 ICS NDK, JNI, Android **services** and LBS.
- Developed Java Context Source **API** and & INTEL CAC framework to implemented as Android **Services**.
- Developed Context Source application using Facebook Graph **API** and INTEL CAC Framework middleware engine and implemented on INTEL Midfield Android ICS phones.
- Developed and built OpenCV libraries in Android NDK r5 and Face-recognition porting in Intel ICS phones.

Environment:

- **Web services** JSON, REST, HTTP
- Android Jelly Bean, ICS, SQLite, Github
- INTEL Android4.0 ICS Midfield Phone
- Android r5 NDK, Location Based **services**, OpenCV
- Android JNI, OpenSurfcpp, Java Context Source
- OpenCV2.4, Facebook Graph **API**, jar libraries.

SPECTRACORE TECHNOLOGIES INC, San Diego, CA

Oct 2011 – Dec 2011

As an Android **Developer** – (Client - KYOCERA COMMN INC.)

- Investigated and Re-implemented Callback Functionality for KYOCERA Milano C5121 Sprint **mobile**.
- Implemented different Android UI menus for Sprint, Movilnet, Page Plus and IUSACell **mobile** service providers.
- Investigated and fixed Emergency Number dialing (911 and 411) issues using Agilent Wireless Commn Sys Tools.
- Implemented MMS network Scale factor (2MB-1MB) for KYOCERA Milano C5121 Sprint service provider.
- Investigated in Resolving link error in JNI/Bionic implementation, integrated various DNS related patches,
- Involved in developing Android UIs and re-implementing, verifying, investigating, finalizing, issues reproducing, troubleshooting, **testing**, debugging, and fixing of CRs from different **mobile** service providers.

PROJECTS HANDLED:

- Developed MCC/MNC hidden debug UI for KYOCERA Milano C5120 Movilnet **mobile** service provider.
- Developed Wi-Fi Tethering UI for KYOCERA Milano C5121 Sprint **mobile** service provider.
- Developed SID/NID system level integration for KYOCERA Milano C5120 Movilnet Mexico service provider.

Environment:

- KYOCERA Milano C5120 and C5121
- Perforce (P4V) – Version Control
- Chameleon - Service Programming
- Qualcomm Tools – QXDM, QPST
- Wire shark – Wi-Fi packet analyzer
- Agilent Wireless Communication Sys.
- Linux-Ubuntu, Eclipse
- Android SDK, NDK, ADT Plug-in, SQLite

PCI Limited, Singapore

Dec 2010 – Apr 2011

As a Senior **Developer** - Android:

- Involved in Android 2.3 Frameworks and **API**'s for Multi Format printer using and OMAP3530.
- Developed QUALCOM DIU50 Android **mobile applications** using common Android framework **API**'s and Android SDK based application **development**.
- Implemented 5 layer Android Bluetooth **applications** using Bluetooth adapter, device, socket, class and server socket.
- Involved in Android board porting and flashing the Android kernel image to i.MX515 EVK board SD Card.
- Developed new features and codes to support application **development** and Utilized UI wireframes and Android tablet.
- Involved in optimization, troubleshooting, **testing** and customer support of various Android projects.

PROJECTS HANDLED:

- Developed Single Board **Computer** solution using Android2.3 frameworks and OMAP3530.



- Personal Content Convergence for Home Network using Samsung Galaxy 10.1 Tablet.
- QUALCOM DIU50 Board level functional tester using free scale i.MX515, Android SDK and Widgets.
- Android Bluetooth Wireless Tester using Free scale i.MX515 and Android framework **API's**.

Environment:

- Samsung Galaxy 10.1 Tablet
- OMAP5330, Samsung ACE **mobile**
- Free Scale i.MX515 EVK board
- Android SDK, NDK, ADT Plug-in, SQLite
- Linux-Ubuntu, Eclipse, QEMU Emulator
- GIT, Android Debugging Bridge-ADB

TRANS INFO TECH PVT LTD, Singapore

MAY 2010 – DEC 2010

As a Senior Engineer - Firmware:

- Implemented all data and UI structures to run on the Android **Mobile** Platform to the client's specifications.
- Constructed a real-time venue rating system for the Android OS allowing cell phone users to get current user ratings for nearby venues using location aware technologies.
- Designed, developed and deployed Android Google map application for Singapore Comfort Delgro Taxis using Friendly ARM mini 2440.
- Responsible for the creating and developing the enterprise application based on J2ME/RIM device.

MAJOR PROJECTS HANDLED:

- Android Location Finder for Singapore Comfort Delgro Taxis using Samsung S3C2440A ARM920T.
- Butterfly Loyalty card and DBS Rebate Redeem and Voucher Redeem Cards.
- EZLink Card (CEPAS) Implementation in Comfort Delgro Taxis.

Environment:

- Android SDK, NDK, ADT Plug-in, SQLit
- Eclipse3.0, Linux-Ubuntu, J2ME
- Friendly ARM mini 2440, OMAP3530
- Java Wireless Toolkit for CLDC
- J2ME **Mobile** apps programming
- Samsung S3C2440A ARM920T

Certifications, Singapore

JAN 2009 – DEC 2009

Microsoft corporations and Sun Microsystems:

- **Microsoft Windows** Embedded CE 6.0 and **Windows Mobile** 5.0.
- Sun Certified Java Programmer for Java Platform, SE 5.0.

Hewlett Packard (hp), Singapore

JAN 2008 – DEC 2008

As a Senior Firmware Engineer:

- Developed WinCE5.0 SDK and OS run time for HP Tester using Free Scale i.MX515 EVK board.
- Developed VS2005 C# application for HP Tester to deploy on Free Scale i.MX515 EVK board via ActiveSync.
- Developed TCP/IP Client Server module **development** with Win32 and MFC console Application.
- Developed C#, C++ and win32 smart device apps, its compiled and debugged successfully using the WinCE6.0.
- Developed Client Server programming using TCP/IP, ISO8583 and VC++ to all POS terminals.

MAJOR PROJECTS HANDLED:

- HP LORENTS based / EDELMAN Final Tester using free scale i.MX515 and WinCE6.0 SDK and OS image.
- Client Server Application Programming Inkjet Printer using VC++, WIN32, MFC **API's**



- Device Driver **Development** for Smart Cards Systems using Multi Thread Socket Programming
- Modern IBM USB Joystick for Flight Control System like Thrust master Joystick
- IBM Rational Robotics and Barcode scanning and Printing system using .NET framework

Environment:

- Win32 **API**, MFC
- Socket Programming, MFC, C#.NET
- Smart Motor, Smart RF Cards
- Embedded C, C++, Visual C++6.0
- Assembly Language program
- WinCE5.0, SDK and OS image
- i.MX515 Ethernet & UART apps
- MCX – Machine Extended Control

PENTAMASTER TECHNOLOGY, Malaysia

JUN 2006 - OCT 2007

As a Senior Engineer - R&D:

- Worked with Embedded C, C++, VC++ for Micro controllers (ARM, ATMEL 89C51, INTEL, PIC) and Assembly, Embedded C, C++, VC++, MFC Programming and GUI with 8bits /16bits/32bits controllers.
- Worked as a talented Senior **Software** Engineer with proven success designing and writing **applications** in Lab VIEW and C/C++ for **Automated** Test Equipment (ATE).
- Designed Zigbee prototype board for 16 channel **applications** using Maxstream Xbee modem.

MAJOR PROJECTS HANDLED:

- Asset-tracking device utilizing GPS positioning technology and GSM cell phone technology.
- PCI data acquisition card and RFID based access control system.
- Open Short Leakage Tester using PLANET ATE chips, Pin Electronics, PMUs, and Multiplexers for semiconductor manufacturers **testing**.
- Advance ECG Monitoring Systems which is mainly used to real-time monitor peoples ECG signal
- ATE System for Semiconductor manufacturing companies

Environment:

- C, C++, Assembly Language
- GUI - MFC, VC++, .NET
- RTOS - WinCE 5.0, WinCE6.0
- Win Socket Level Programming
- Serial commn., Bluetooth USB, TCP/IP
- PLANET ATE Chips, Pin Elexs PMU
- ARM9 Samsung SMDK2443
- Smart Motors – 16 bits Panasonic
- Smart Cards – IC Chips, MAG strips
- Smart Sensors - Bar Code, IR, Wi-Fi

APPLIED AUTOMATION SYSTEMS PVT LTD, India

MAY 2005 – JUN 2006

As a Senior Engineer – R&D

- Worked in programming with Assembly Language for PIC18F4550 using MPASM assembler and EMBEDDED C, C++ Programming for PIC18F4550 using MCC18 Compiler.
- Involved in Prototype **development** with CAN2.0 using PIC12C672 & MCP2515, MCP2551 and Programming Knowledge with CAPL – CAN Access Programming Language.
- Involved in Prototype hardware **design** and firmware (C and assembler) **development** of an Intel 8051 8-bit microcontroller based multi-channel motor speed sensor incorporating networked serial optical sensors.

MAJOR PROJECTS HANDLED:

- Robotics Motion Control project and 30-axis motion control system.
- Robotics Motion Control System for Injection Molding systems
- Multi Axis (30-Axis) Servo Motion Control System for Semiconductor **Testing** Systems



VETAL TEXTILES AND ELECTRONICS PVT LTD, India

APR 1997 – MAY 2005

As a Senior Engineer – R&D

- **Design** and Developed Intelligent display systems (IDS) to world class client Lakshmi Machine Works (LMW) Limited using Multiprocessor, Embedded C, C++ and Visual Basic Platforms.
- Designed the RS-232 **interface** to PC and defined the handshake protocol for communicating to PC **software**, devised and implemented user tunable digital filters in assembly **software**.
- Designed and developed the signal conditioning for weather sensors, the A/D converter using MCP3202 (2 channel) and MCP3204 (4 channel) and microprocessor modules.
- Created script files to automate compilation and linking of source files using IAR C-compiler.

MAJOR PROJECTS HANDLED:

- Microcontroller based Metal Detector and Spark detectors using Multi Processor.
- Cotton Contamination Machine with Poly Sensor using CCD Camera.
- Universal Color Sorter detection machine for modern rice milling machinery plant.
- Attendance Recorder and Security Register system.
- Online Charger Synchronizer (for Southern – Indian - railways).

Environment:

- ATMEL 89C51, 89C1051, 89C2051
- RS232, RS485, SPI, I2C, Code Composer
- ADC, DAC, EEPROM, Line scan camera
- Kiel C, MCC18, ICD2 MPLAB, MPASM
- Assembly, Embedded C, C++
- VC++, Visual Basic
- Microchip 16F876, TI MSP430
- PCB **design** – PROTEL, ORCAD

PROFESSIONAL CERTIFICATIONS

Year	Certification	Certified By
SEP 2022	AWS Certified Cloud Practitioner	https://www.credly.com/badges/f64c2842-6cf4-4be0-a463-32b8854c890d
APRIL 2023	AWS Certified Solutions Architect	https://www.credly.com/badges/ba2a3b18-2561-45f5-8be2-98b2576ad97b
JUNE 2022	AWS Certified DevOps Engineer	Amazon - AWS
JULY 2022	AWS Certified Machine Learning	Amazon - AWS
Oct 2009	Microsoft Windows Embedded CE 6.0, Development	Microsoft Corporation
Nov 2009	Microsoft Windows Mobile 5.0 - Application Development	Microsoft Corporation
Dec 2009	ASP.NET & Web Developer by Using Microsoft .NET FW 3.5	Microsoft Corporation
Dec 2009	Sun Certified Java Programmer for Java Platform, SE 5.0	Sun Microsystems
Mar 2011	Embedded Android Development Training	International SQL Star
Jul 2011	Oracle Certified Professional, Java ME 1 Mobile Application	Oracle University



EDUCATION

Bachelors of Engineering, Anna University – INDIA.

Diploma in Electronics and communication Engineering, Mohammed Sathak Polytechnic, Tamil Nadu, India.

Anbalagan Lakshmanan

- Irving, TX, US

Contact Information

- 164-ku5-w1x@mail.dice.com
- 7132810039

Summary

Mobile (Android and iOS) Application Architect with over 17 years of total **Software development** experience, which includes 6+ years experiences with integrating Android framework **development** (AOSP), Android Algorithms (Face Detection, Face Recognition), PhoneGap, OpenGL, Dagger 12, Retrofit okHttp, Butterknife, RxJava RxAndroid, Robolectric Junit - for BehaviorTest driven **development** (BDDTDD), Realm Content Provider and Speech Recognition. Technologies: Android SDK OreoNougat Marshmallow Lollipop Kit-Kat , Android JNI and NDK, JAVA, Embedded C, C++, SQLite3 Database, Android **API** s, AOSP, Google Map **API**'s, LBS - Locale3.x, Qualcomm Gimbal, SVN - Versions, Git Gerrit GitRepo, Webservices using XML, REST and JSON. 5+ years **development** and integration experiences with Android **Cloud** Application **Development** includes **Amazon Web Services** (AWS), AWS SNS simple notifications service, AWS Pin Point **mobile** push notifications, AWS Cognito to controls user authentication and access, **Microsoft** MSAL **Azure** B2C and Forgerock **mobile** app integrations to identity and access management. Android **Applications** published in Google Play <https://play.google.com/store/apps/details?id=com.anbu.ringtonemaker&hl=en>

Work History

Total Work Experience: 15 years

- **Lead Android Developer** Newway Healthcare Solutions
Aug 01, 2022
- **Lead Android Developer** Herringbank
Jan 01, 2020

- **Lead Android Developer** Tuya Technology
Jun 01, 2019
- **Lead Android Developer** Toyota Motors North America
May 01, 2018
- **Lead Android Developer** Bank Of America
Aug 01, 2016
- **Senior Mobile Engineer - Android** Intel
Jun 01, 2015
- **Application Architect - Android** American Express
Jun 01, 2014
- **Senior Android Developer** Deutsche Telekom And Freelance **Android Developer** (Sensoplex)
Jan 01, 2014
- **Senior Android Developer** Etouch Inc (Client - Google Mountain View)
Sep 01, 2013
- **Senior Android Developer** Datacard Group
Apr 01, 2013
- **Senior Android Developer** Okaya Inc (Client - Intel, Santa Clara, CA)
Mar 01, 2012
- **Senior Android Developer** Spetracore Technology
Sep 01, 2011
- **Senior Android Developer** PCI LTD, Singapore
Dec 01, 2010

Education

- **Bachelors** | Anna University

Skills

- **constraint layout** - 5 years
- **hilt inject viewmodel** - 5 years
- **jetpack view binding** - 5 years
- **kotlin coroutines and multithreading** - 5 years
- **kotlin synthetics** - 5 years
- **livedata** - 5 years
- **paging, android architecture components** - 5 years
- **recyclerview** - 5 years
- **use coroutines with livedata** - 5 years
- **amazon web services**
- **anaconda**
- **android**
- **android design**
- **android development**
- **android espresso**
- **android jetpack**
- **android ndk**
- **android sdk**
- **android studio**
- **aosp**
- **architects**
- **authentication protocols**
- **aws cloudformation**
- **aws command line interface**
- **aws identity and access management**
- **aws lambda**
- **bitbucket**
- **c**
- **c++**
- **cascading style sheets**
- **clean architecture**
- **code review**
- **continuous integration and continuous**
- **convolutional neural networks**
- **core java**
- **cucumber**
- **debugging**
- **deep learning**
- **dependency injection**
- **device drivers**
- **domain architecture**
- **eclipse**

- embedded linux
- embedded **software**
- embedded systems
- fastlane
- forgerock identity platform
- generative ai
- gerrit
- git
- github
- google analytics
- google **cloud** platform
- gradle
- hilt
- html5
- ios **development**
- java
- jenkins
- jsp
- keras
- kotlin
- kotlin coroutines
- large language model operations
- large language models
- leading **development** teams
- machine learning
- machine learning algorithms
- **microsoft** app center
- **mobile applications**
- model view viewmodel
- natural language processing
- nlp
- nltk
- numpy
- opencv
- optical character recognition
- pandas
- python
- pytorch
- react native
- representational state transfer
- rest
- rest **api**
- rest apis
- retrofit
- rtos

- rxjava
- scikit learn
- servlets
- software development
- solid design principles
- solution architecture
- sqlite
- subversion
- supervised learning
- swift
- tensorflow
- the microsoft authentication library
- unsupervised learning
- web development
- web services
- xcode
- xml
- android12.0, aosp, jetpack compose, - 9 years
- cdc: jira, jenkins, bitbucket, - 9 years
- dagger, retrofit, okhttp, rxjava, - 9 years
- analytics : transmit sdk, ensighton, - 8 years
- android architect mvvm, mvc experience - 8 years
- c and c++ - 8 years
- sdlc - agile and waterfall - 8 years
- android application and framework - 7 years
- android fragments, binder ipc's - 7 years
- android sdk, ndk development for - 7 years
- android studio 3.4.1 and gradle - 7 years
- bluetooth ble - 7 years
- java1.8 programming - 7 years
- roomdb, mangodb and sql lite - 7 years
- sourcetree, subversions and git - 7 years
- google ad mob - 6 years
- gradle5.3.0, - 6 years
- otp - twilio, amazon pinpoint, - 6 years
- kotlin1.8.0, coroutine - 5 years
- microsoft azure, aws, google cloud - 5 years
- ocr : microblink, ideology, misnap, - 5 years
- rest and json, gson, okhttp, retrofit, - 5 years
- flutter, android native, hybrid - 2 years
- html4 and html5 - 2 years
- javascript - 2 years
- fountain and truckright - 1 years
- opengl, phonegap - 1 years

Work Preferences

- Desired Work Settings: Remote or On-Site or Hybrid
- Likely to Switch: True
- Willing to Relocate: True
- Travel Preference: 100%
- Preferred Location:
 - Houston, TX, US
 - Austin, TX, US
 - San Antonio, TX, US
- Work Authorization:
 - US
- Work Documents:
 - US Citizen
- Security Clearance: False
- Desired Hourly Rate: 80+ (USD)
- Desired Salary: 145,000+ (USD)
- Third Party: False
- Employment Type:
 - Full-time
 - Part-time
 - Contract - Corp-to-Corp
 - Contract - Independent
 - Contract - W2
 - Contract to Hire - Corp-to-Corp
 - Contract to Hire - Independent
 - Contract to Hire - W2

Profile Sources

- linkedin: https://www.linkedin.com/in/anbalagan-lakshmanan-38065518?trk=nav_responsive_tab_profile
- facebook: <https://facebook.com/alex.laxman.12>
- twitter: <https://twitter.com/anbalagan2014>
- gravatar: <https://gravatar.com/anbulaxmanan>
- klout: <https://klout.com/anbalagan2014>
- Dice:
<https://www.dice.com/employer/talent/profile/096a52c72329c70787cae43b341fd3ec>